

Problem 1 [靜態模型：景氣波動] (21%)

考慮一個靜態的經濟體：代表性消費者的偏好為 $u = \log c + \log l$ ，其中， c 為商品需求、 l 為休閒；生產函數為 $y = zn^{1/2}$ ，其中， y 為商品供給、 z 為總要素生產力、 n 為勞動。家戶的時間稟賦為 $T = 3$ 。

(1) 求出經濟體的中央規畫者的解 (central planner solution)。(7%)

$$\begin{aligned} \max_{c, l} \quad & u = \log c + \log l \\ \text{s.t.} \quad & c = y = zn^{1/2} \\ & n + l = 3 \end{aligned}$$

$$\begin{aligned} \text{FOC} \quad [n] \quad & \frac{1}{c} \left(\frac{1}{2} zn^{-1/2} \right) + \frac{1}{l} (-1) = 0 \\ \rightarrow \quad & \frac{z}{c} (n^{-1/2}) = \frac{2}{l} \\ \rightarrow \quad & \frac{z}{zn^{1/2}} (n^{-1/2}) = \frac{2}{3-n} \\ \rightarrow \quad & \frac{1}{n} = \frac{2}{3-n} \\ \rightarrow \quad & n = 1 \\ & l = 2 \\ & c = y = z \end{aligned}$$

(2) 描述競爭均衡 (competitive equilibrium)。以圖與函數標示 (a) 勞動需求, (b) 勞動供給, (c) 勞動市場均衡。(7%)

$$\begin{aligned} \max_n \quad & \pi = y - wn \\ \text{s.t.} \quad & y = zn^{1/2} \\ \text{FOC} \quad [n] \quad & \frac{1}{2} zn^{-1/2} - w = 0 \\ \rightarrow \quad & n^{1/2} = \frac{z}{2w} \\ \rightarrow \quad & n = \frac{z^2}{4w^2} = n^d \\ & y = \frac{z^2}{2w} \\ & \pi = \frac{z^2}{4w} \end{aligned}$$

$$\begin{aligned} \max_{c, l} \quad & u = \log c + \log l \\ \text{s.t.} \quad & c = wn + \pi \\ & n + l = 3 \\ \text{FOC} \quad [n] \quad & \frac{1}{c}(w) + \frac{1}{l}(-1) = 0 \\ \rightarrow \quad & \frac{1}{c}(w) = \frac{1}{l} \\ \rightarrow \quad & c = wl \\ \rightarrow \quad & l = \frac{3}{2} + \frac{\pi}{2w} \\ & n = \frac{3}{2} - \frac{\pi}{2w} = n^s \\ & c = \frac{3w + \pi}{2} \end{aligned}$$

$$\begin{aligned}
 \text{Labor market equilibrium} \quad n^s &= n^d \\
 &\rightarrow \frac{3}{2} - \frac{\pi}{2w} = \frac{z^2}{4w^2} \\
 \pi = \frac{z^2}{4w} &\rightarrow \frac{3}{2} - \frac{z^2}{8w^2} = \frac{z^2}{4w^2} \\
 &\rightarrow \frac{3}{2} = \frac{3z^2}{8w^2} \\
 &\rightarrow w^2 = \frac{z^2}{4} \\
 &\rightarrow w = \frac{z}{2} \\
 &\rightarrow n = 1 \\
 &\quad l = 2 \\
 &\quad c = y = z \\
 &\quad \pi = \frac{z^2}{4(z/2)} = \frac{z}{2}
 \end{aligned}$$

(3) 承接第(2)小題。假設經濟體原處於競爭均衡，現因某些非理性因素使得市場疑慮湧現，商品需求降為 $c = z/2$ 。描述勞動市場在 (a)工資維持原水準時, (b)工資可充分調整時的結果。(7%)

(a) 工資維持原水準： $w = \frac{z}{2}$

$$\text{廠商} \quad n = \frac{z^2}{4w^2} \rightarrow n^d = \frac{z^2}{4(z^2/4)} = 1$$

$$\begin{aligned}
 \text{家戶} \quad c = wl &\rightarrow \left(\frac{z}{2}\right) = \left(\frac{z}{2}\right) \cdot l \\
 &\rightarrow l = 1 \\
 &\rightarrow n^s = 2
 \end{aligned}$$

勞動市場不均衡

(a) 工資可充分調整： w'

商品市場： $y^* = c^* = z/2$

廠商

$$y = \frac{z^2}{2w} \rightarrow \frac{z}{2} = \frac{z^2}{2w} \rightarrow w' = z$$

$$n = \frac{z^2}{4w^2} \rightarrow n = \frac{z^2}{4(z)^2} = \frac{1}{4}$$

Problem 2 [稟賦經濟：消費稅與李嘉圖等值] (28%)

考慮一個兩期的稟賦經濟模型。消費者的效用為 $u = \log c_1 + \frac{\log c_2}{1+\rho}$ ，稟賦為 (y_1, y_2) 。政府對消費者兩期消費分別課徵稅率為 τ_1 與 τ_2 的消費稅。政府的兩期支出為 (g_1, g_2) 。政府的兩期預算限制分別為

$$g_1 = \tau_1 c_1 + b_1^g,$$

$$g_2 + (1+r)b_1^g = \tau_2 c_2.$$

(1) 寫出消費者的極大化問題，求解一階條件(Euler equations) 及最適消費函數。(7%)

$$\max_{c_1, c_2} \quad u = \log c_1 + \frac{\log c_2}{1+\rho}$$

$$\text{s.t} \quad (1+\tau_1)c_1 + \frac{(1+\tau_2)c_2}{1+r} = y_1 + \frac{y_2}{1+r}$$

$$\text{FOC} \quad u'(c_1) = \frac{1}{1+\rho} \cdot u'(c_2) \cdot \frac{(1+\tau_1)}{(1+\tau_2)}(1+r)$$

$$\rightarrow \quad \frac{1}{c_1} = \frac{1}{1+\rho} \cdot \frac{1}{c_2} \cdot \frac{(1+\tau_1)}{(1+\tau_2)}(1+r)$$

利用 FOC 與限制式，求出 c_1, c_2 。也可利用公式直接寫出：

$$\rightarrow \quad c_1 = \frac{1+\rho}{2+\rho} \cdot \left(y_1 + \frac{y_2}{1+r} \right) \cdot \frac{1}{1+\tau_1}$$

$$\rightarrow \quad c_2 = \frac{1}{2+\rho} \cdot \left(y_1 + \frac{y_2}{1+r} \right) \cdot \frac{1+r}{1+\tau_2}$$

(2) 計算經濟體均衡時的利率 r^* 。以外生變數和參數表示。(7%)

$$\text{FOC (Euler equation)} \quad \frac{1}{c_1} = \frac{1}{1+\rho} \cdot \frac{1}{c_2} \cdot \frac{(1+\tau_1)}{(1+\tau_2)}(1+r) \quad (1)$$

$$\text{HH's constraints} \quad (1+\tau_1)c_1 + b_1^g = y_1 \quad (2)$$

$$(1+\tau_2)c_2 = y_2 + (1+r)b_1^g \quad (3)$$

$$\text{GOV's constraints} \quad g_1 = \tau_1 c_1 + b_1^g \quad (4)$$

$$g_2 + (1+r)b_1^g = \tau_2 c_2 \quad (5)$$

$$\text{market-clear conditions} \quad c_1 + g_1 = y_1 \quad (6)$$

$$c_2 + g_2 = y_2 \quad (7)$$

利用 (2),(3),(4),(5)，可得 (6)(7)。

利用 (1),(6),(7)，將 $c_1 = y_1 - g_1$ 、 $c_2 = y_2 - g_2$ 代換：

$$\rightarrow \quad \frac{1}{(y_1 - g_1)} = \frac{1}{1+\rho} \cdot \frac{1}{(y_2 - g_2)} \cdot \frac{(1+\tau_1)}{(1+\tau_2)}(1+r)$$

$$\rightarrow \quad 1+r = (1+\rho) \cdot \frac{(1+\tau_2)}{(1+\tau_1)} \cdot \frac{(y_2 - g_2)}{(y_1 - g_1)}$$

$$\rightarrow \quad r = (1+\rho) \cdot \frac{(1+\tau_2)}{(1+\tau_1)} \cdot \frac{(y_2 - g_2)}{(y_1 - g_1)} - 1$$

- (3) 選舉當前，民主黨政府 (Democrats are donkeys.) 考慮赤字減稅 (τ_1 下降) 政策，並改以增加舉債 (b_1^g 上升) 因應。其他條件不變，說明市場均衡利率的變化情形。對民眾而言，是否有影響？以繪圖、數式或文字等方式說明。(7%)

$$r^* = (1 + \rho) \cdot \frac{(1 + \tau_2)}{(1 + \tau_1)} \cdot \frac{(y_2 - g_2)}{(y_1 - g_1)} - 1$$

$$\rightarrow \frac{\partial r^*}{\partial \tau_1} < 0 \text{ 即稅率 } \tau_1 \text{ 下降、均衡利率 } r^* \text{ 上升。}$$

$$u = \log c_1 + \frac{\log c_2}{1 + \rho} = \log(y_1 - g_1) + \frac{\log(y_2 - g_2)}{1 + \rho}$$

只要政府支出 g_1, g_2 不變，則民間消費 $c_1 = y_1 - g_1$ 與 $c_2 = y_2 - g_2$ 皆不變。

家戶的效用不變。減稅政策對民眾無任何影響。

- (4) 信奉小政府理念的共和黨經濟學家 (Republicans are elephants.) 認為只有減少政府開銷 (g_1, g_2 下降) 才有助於提升民眾福祉。此論述是否正確？是否有前提存在？以繪圖、數式或文字等說明之。(7%)

$$u = \log c_1 + \frac{\log c_2}{1 + \rho} = \log(y_1 - g_1) + \frac{\log(y_2 - g_2)}{1 + \rho}$$

政府支出 g_1, g_2 減少，則民間消費 $c_1 = y_1 - g_1$ 與 $c_2 = y_2 - g_2$ 增加。

家戶的效用提升！

Problem 3 [跨期勞動選擇] (28%)

考慮一個兩期 ($t = 1, 2$) 的經濟體，包含廠商部門 (Firms) 與家戶部門 (Households)。廠商雇用勞動進行生產並支付勞動薪資。家戶提供勞動以獲取薪資收入，且家戶為廠商之股東，領取廠商之獲利。家戶可同時進行儲蓄或借貸的決策，實質利率為 r 。實質薪資率為 w_t 。

廠商的生產函數為 $y_t = f(n_t) = A_t n_t^\alpha$ 、利潤為 $\pi_t = y_t - w_t n_t$ 。廠商追求利潤極大。家戶的效用函數為

$$u = \sum_{t=1}^2 \beta^{t-1} (\ln c_t + \ln l_t), \text{ 且時間稟賦為 } 1。 \text{ 其餘模型設定，與課程內容一致。}$$

- (1) 寫出廠商的極值問題 (firms' problem)：必須包含目標函數、限制式與一階條件。(7%)

$$\max_{n_1, n_2} \quad \pi_1 + \frac{\pi_2}{1+r}$$

$$\max_{n_1, n_2} \quad (y_1 - w_1 n_1) + \frac{(y_2 - w_2 n_2)}{1+r}$$

$$\text{s.t.} \quad y_1 = A_1 n_1^\alpha$$

$$y_2 = A_2 n_2^\alpha$$

$$\text{FOC} \quad [n_1] \quad \alpha A_1 n_1^{\alpha-1} = w_1$$

$$[n_2] \quad \alpha A_2 n_2^{\alpha-1} = w_2$$

- (2) 寫出家戶的極值問題 (households' problem)：必須包含目標函數、限制式與所有一階條件。(7%)

$$\max_{c_1, c_2, l_1, l_2} \quad u = (\ln c_1 + \ln l_1) + \beta (\ln c_2 + \ln l_2)$$

$$\text{s.t.} \quad c_1 + b_1 = w_1 n_1 + \pi_1$$

$$c_2 = w_2 n_2 + \pi_2 + (1+r)b_1$$

$$\rightarrow \quad (w_1 l_1 + c_1) + \frac{(w_2 l_2 + c_2)}{1+r} = (w_1 + \pi_1) + \frac{(w_2 + \pi_2)}{1+r}$$

$$\text{FOC} \quad u'(c_1) = \beta u'(c_2)(1+r) \quad \rightarrow \quad \frac{1}{c_1} = \beta \frac{1}{c_2}(1+r)$$

$$u'(l_1) = u'(c_1)(w_1) \quad \rightarrow \quad \frac{1}{l_1} = \frac{1}{c_1}(w_1)$$

$$u'(l_2) = u'(c_2)(w_2) \quad \rightarrow \quad \frac{1}{l_2} = \frac{1}{c_2}(w_2)$$

$$u'(l_1) = \beta u'(l_2) \frac{w_1(1+r)}{w_2} \quad \rightarrow \quad \frac{1}{l_1} = \beta \frac{1}{l_2} \frac{w_1(1+r)}{w_2}$$

(3) 求出一般均衡 (general equilibrium)：必須包含所有內生變數的解。(7%)

FOC

$$\alpha A_1 n_1^{\alpha-1} = w_1$$

$$\alpha A_2 n_2^{\alpha-1} = w_2$$

$$\frac{1}{c_1} = \beta \frac{1}{c_2} (1+r)$$

$$\frac{1}{l_1} = \beta \frac{1}{l_2} \frac{w_1(1+r)}{w_2}$$

$$\frac{1}{l_1} = \frac{1}{c_1} (w_1)$$

$$\frac{1}{l_2} = \frac{1}{c_2} (w_2)$$

market-clearing conditions

$$c_1 = y_1 = w_1 n_1 + \pi_1$$

$$c_2 = y_2 = w_2 n_2 + \pi_2$$

$$w_1 n_1 = \alpha y_1$$

$$\pi_1 = (1-\alpha)y_1$$

$$w_2 n_2 = \alpha y_2$$

$$\pi_2 = (1-\alpha)y_2$$

利用同期條件解 w

$$\frac{1}{l_1} = \frac{1}{c_1} (w_1)$$

$$\rightarrow \frac{1}{1-n_1} = \frac{1}{y_1} \left(\frac{\alpha y_1}{n_1} \right)$$

$$\rightarrow \frac{1}{1-n_1} = \frac{\alpha}{n_1}$$

$$\rightarrow n_1 = \frac{\alpha}{1+\alpha}$$

$$\rightarrow y_1 = A_1 \left(\frac{\alpha}{1+\alpha} \right)^\alpha$$

$$\rightarrow w_1 = \alpha A_1 \left(\frac{\alpha}{1+\alpha} \right)^{\alpha-1}$$

同理可得

$$n_2 = \frac{\alpha}{1+\alpha}$$

$$\rightarrow y_2 = A_2 \left(\frac{\alpha}{1+\alpha} \right)^\alpha$$

$$\rightarrow w_2 = \alpha A_2 \left(\frac{\alpha}{1+\alpha} \right)^{\alpha-1}$$

利用跨期條件解 r

$$\frac{1}{c_1} = \beta \frac{1}{c_2} (1+r)$$

$$\rightarrow \frac{1}{y_1} = \beta \frac{1}{y_2} (1+r)$$

$$\rightarrow 1+r = \frac{1}{\beta} \cdot \frac{y_2}{y_1}$$

$$\rightarrow 1+r = \frac{1}{\beta} \cdot \frac{A_2 \left(\frac{\alpha}{1+\alpha} \right)^\alpha}{A_1 \left(\frac{\alpha}{1+\alpha} \right)^\alpha}$$

$$\rightarrow 1+r = \frac{1}{\beta} \cdot \frac{A_2}{A_1}$$

$$\rightarrow r = \frac{1}{\beta} \cdot \frac{A_2}{A_1} - 1$$

整理

$$n_1 = \frac{\alpha}{1 + \alpha}$$

$$l_1 = \frac{1}{1 + \alpha}$$

$$c_1 = A_1 \left(\frac{\alpha}{1 + \alpha} \right)^\alpha$$

$$y_1 = A_1 \left(\frac{\alpha}{1 + \alpha} \right)^\alpha$$

$$w_1 = \alpha A_1 \left(\frac{\alpha}{1 + \alpha} \right)^{\alpha-1}$$

$$n_2 = \frac{\alpha}{1 + \alpha}$$

$$l_2 = \frac{1}{1 + \alpha}$$

$$c_2 = A_2 \left(\frac{\alpha}{1 + \alpha} \right)^\alpha \quad y_2 = A_2 \left(\frac{\alpha}{1 + \alpha} \right)^\alpha \quad w_2 = \alpha A_2 \left(\frac{\alpha}{1 + \alpha} \right)^{\alpha-1}$$

$$r = \frac{1}{\beta} \cdot \frac{A_2}{A_1} - 1$$

- (4) 假設政府於第一期課徵勞動所得稅，稅率為 t_1 。並且將稅收投資於債券市場，於第二期全數補貼勞動薪資，補貼率為 s_2 。政府的收支方程式為 $t_1 w_1 n_1 \cdot (1+r) = s_2 w_2 n_2$ 。與第(3)題相比較，均衡時兩期的薪資與勞動量有何變化？(7%)

廠商部門未有任何變動

家戶部門更改為

$$\begin{aligned} \max_{c_1, c_2, l_1, l_2} \quad & u = (\ln c_1 + \ln l_1) + \beta (\ln c_2 + \ln l_2) \\ \text{s.t.} \quad & c_1 + b_1 = (1 - t_1) w_1 n_1 + \pi_1 \\ & c_2 = (1 + s_2) w_2 n_2 + \pi_2 + (1 + r) b_1 \\ \rightarrow \quad & [(1 - t_1) w_1 l_1 + c_1] + \frac{[(1 + s_2) w_2 l_2 + c_2]}{1 + r} = [(1 - t_1) w_1 + \pi_1] + \frac{[(1 + s_2) w_2 + \pi_2]}{1 + r} \end{aligned}$$

$$\begin{aligned} \text{FOC} \quad & u'(c_1) = \beta u'(c_2)(1 + r) \quad \rightarrow \quad \frac{1}{c_1} = \beta \frac{1}{c_2}(1 + r) \\ & u'(l_1) = u'(c_1)(w_1) \quad \rightarrow \quad \frac{1}{l_1} = \frac{1}{c_1}[(1 - t_1) w_1] \\ & u'(l_2) = u'(c_2)(w_2) \quad \rightarrow \quad \frac{1}{l_2} = \frac{1}{c_2}[(1 + s_2) w_2] \\ & u'(l_1) = \beta u'(l_2) \frac{w_1(1 + r)}{w_2} \quad \rightarrow \quad \frac{1}{l_1} = \beta \frac{1}{l_2} \frac{(1 - t_1) w_1(1 + r)}{(1 + s_2) w_2} \end{aligned}$$

一般均衡 (general equilibrium)

$$\begin{aligned} \text{FOC} \quad & \alpha A_1 n_1^{\alpha-1} = w_1 & \alpha A_2 n_2^{\alpha-1} = w_2 \\ & \frac{1}{c_1} = \beta \frac{1}{c_2}(1 + r) & \frac{1}{l_1} = \beta \frac{1}{l_2} \frac{(1 - t_1) w_1(1 + r)}{(1 + s_2) w_2} \\ & \frac{1}{l_1} = \frac{1}{c_1}[(1 - t_1) w_1] & \frac{1}{l_2} = \frac{1}{c_2}[(1 + s_2) w_2] \\ \text{market-clearing conditions} \quad & c_1 = y_1 = w_1 n_1 + \pi_1 & c_2 = y_2 = w_2 n_2 + \pi_2 \\ & w_1 n_1 = \alpha y_1 & \pi_1 = (1 - \alpha) y_1 \\ & w_2 n_2 = \alpha y_2 & \pi_2 = (1 - \alpha) y_2 \end{aligned}$$

利用同期條件解 w

$$\begin{aligned}
\frac{1}{l_1} &= \frac{1}{c_1}[(1-t_1)w_1] & \rightarrow & \frac{1}{1-n_1} = \frac{1}{y_1}[(1-t_1)(\frac{\alpha y_1}{n_1})] \\
& & \rightarrow & \frac{1}{1-n_1} = (1-t_1)(\frac{\alpha}{n_1}) \\
& & \rightarrow & n_1 = \frac{\alpha(1-t_1)}{1+\alpha(1-t_1)} \\
& & \rightarrow & y_1 = A_1 \left(\frac{\alpha(1-t_1)}{1+\alpha(1-t_1)} \right)^\alpha \\
& & \rightarrow & w_1 = \alpha A_1 \left(\frac{\alpha(1-t_1)}{1+\alpha(1-t_1)} \right)^{\alpha-1}
\end{aligned}$$

利用同期條件解 w

$$\begin{aligned}
\frac{1}{l_2} &= \frac{1}{c_2}[(1+s_2)w_2] & \rightarrow & \frac{1}{1-n_2} = \frac{1}{y_2}[(1+s_2)(\frac{\alpha y_2}{n_2})] \\
& & \rightarrow & \frac{1}{1-n_2} = (1+s_2)(\frac{\alpha}{n_2}) \\
& & \rightarrow & n_2 = \frac{\alpha(1+s_2)}{1+\alpha(1+s_2)} \\
& & \rightarrow & y_2 = A_2 \left(\frac{\alpha(1+s_2)}{1+\alpha(1+s_2)} \right)^\alpha \\
& & \rightarrow & w_2 = \alpha A_2 \left(\frac{\alpha(1+s_2)}{1+\alpha(1+s_2)} \right)^{\alpha-1}
\end{aligned}$$

利用跨期條件解 r

$$\begin{aligned}
\frac{1}{c_1} &= \beta \frac{1}{c_2}(1+r) & \rightarrow & \frac{1}{y_1} = \beta \frac{1}{y_2}(1+r) \\
& & \rightarrow & 1+r = \frac{1}{\beta} \cdot \frac{y_2}{y_1} \\
& & \rightarrow & 1+r = \frac{1}{\beta} \cdot \frac{A_2 \left(\frac{\alpha(1+s_2)}{1+\alpha(1+s_2)} \right)^\alpha}{A_1 \left(\frac{\alpha(1-t_1)}{1+\alpha(1-t_1)} \right)^\alpha} \\
& & \rightarrow & r = \frac{1}{\beta} \cdot \frac{A_2 \left(\frac{\alpha(1+s_2)}{1+\alpha(1+s_2)} \right)^\alpha}{A_1 \left(\frac{\alpha(1-t_1)}{1+\alpha(1-t_1)} \right)^\alpha} - 1
\end{aligned}$$

整理

$$n_1 = \frac{\alpha(1-t_1)}{1+\alpha(1-t_1)}$$

$$c_1 = y_1 = A_1 \left(\frac{\alpha(1-t_1)}{1+\alpha(1-t_1)} \right)^\alpha$$

$$w_1 = \alpha A_1 \left(\frac{\alpha(1-t_1)}{1+\alpha(1-t_1)} \right)^{\alpha-1}$$

$$n_2 = \frac{\alpha(1+s_2)}{1+\alpha(1+s_2)}$$

$$c_2 = y_2 = A_2 \left(\frac{\alpha(1+s_2)}{1+\alpha(1+s_2)} \right)^\alpha$$

$$w_2 = \alpha A_2 \left(\frac{\alpha(1+s_2)}{1+\alpha(1+s_2)} \right)^{\alpha-1}$$

$$r = \frac{1}{\beta} \cdot \frac{A_2 \left(\frac{\alpha(1+s_2)}{1+\alpha(1+s_2)} \right)^\alpha}{A_1 \left(\frac{\alpha(1-t_1)}{1+\alpha(1-t_1)} \right)^\alpha} - 1$$

第(4)小題 v.s. 第(3)小題

$$n_1 = \frac{\alpha(1-t_1)}{1+\alpha(1-t_1)}$$

<

$$n_1 = \frac{\alpha}{1+\alpha}$$

$$w_1 = \alpha A_1 \left(\frac{\alpha(1-t_1)}{1+\alpha(1-t_1)} \right)^{\alpha-1}$$

>

$$w_1 = \alpha A_1 \left(\frac{\alpha}{1+\alpha} \right)^{\alpha-1}$$

$$n_2 = \frac{\alpha(1+s_2)}{1+\alpha(1+s_2)}$$

>

$$n_2 = \frac{\alpha}{1+\alpha}$$

$$w_2 = \alpha A_2 \left(\frac{\alpha(1+s_2)}{1+\alpha(1+s_2)} \right)^{\alpha-1}$$

<

$$w_2 = \alpha A_2 \left(\frac{\alpha}{1+\alpha} \right)^{\alpha-1}$$

Problem 4 [Ramsey and Growth] (23%)

考慮以下的 Ramsey 模型：

效用函數為 $\sum_{t=1}^n \beta^{t-1} u(c_t)$ ；生產函數為 $y_t = A_t f(k_t)$ ；資本累積式為 $k_{t+1} = (1 - \delta)k_t + i_t$ 。跨期的實質利率為 r_t 。

經濟體的市場結清條件為 $c_t + i_t = y_t$ 。

(1) 寫出完整的效用極大模型：無窮期且包含各期限限制條件。(7%)

$$\begin{aligned} \max_{c_t} \quad & \sum_{t=1}^{\infty} \beta^{t-1} u(c_t) \\ \text{s.t.} \quad & y_t = A_t f(k_t) \\ & k_{t+1} = (1 - \delta)k_t + i_t \end{aligned}$$

$$\begin{aligned} \max_{c_1, c_2, c_3, \dots} \quad & u(c_1) + \beta u(c_2) + \beta^2 u(c_3) + \dots \\ \text{s.t.} \quad & c_1 + i_1 = y_1 \\ & c_2 + i_2 = y_2 \\ & c_3 + i_3 = y_3 \\ & \dots \\ & y_1 = A_1 f(k_1) \\ & y_2 = A_2 f(k_2) \\ & y_3 = A_3 f(k_3) \\ & \dots \\ & k_2 = (1 - \delta)k_1 + i_1 \\ & k_3 = (1 - \delta)k_2 + i_2 \\ & k_4 = (1 - \delta)k_3 + i_3 \end{aligned}$$

- (2) 計算兩期 ($t = 1, 2$) 模型的均衡解：消費 c_1^*, c_2^* 與投資 i_1^*, i_2^* 為何？均衡實質利率 r^* 為何？假設經濟體的資訊如下： $u(c_t) = \ln c_t$ 、 $y_t = A_t k_t^\alpha$ 、 $\alpha = 1$ 、 $A_1 = A_2 = 1$ 、 $\beta = 1$ 、 $\delta = 1$ 。已知資本初始條件 $k_1 = k$ 。(8%)

$$\max_{c_1, c_2} \ln c_1 + \beta \ln c_2$$

$$\text{s.t.} \quad c_1 + k_2 = A_1 k_1^\alpha$$

$$c_2 = A_2 k_2^\alpha$$

帶入本題參數

$$\max_{c_1, c_2} \ln c_1 + \beta \ln c_2$$

$$\text{s.t.} \quad c_1 + k_2 = k^\alpha$$

$$c_2 = k_2^\alpha$$

$$\text{FOC} \quad [k_2] \quad \frac{1}{c_1} = \frac{\beta}{c_2} (\alpha k_2^{\alpha-1})$$

$$\rightarrow c_2 = \beta (\alpha k_2^{\alpha-1}) c_1$$

$$c_2 = k_2^\alpha \rightarrow k_2^\alpha = \beta (\alpha k_2^{\alpha-1}) c_1$$

$$\rightarrow k_2 = \alpha \beta c_1$$

$$c_1 + k_2 = k^\alpha \rightarrow c_1 + \alpha \beta c_1 = k^\alpha$$

$$\rightarrow (1 + \alpha \beta) c_1 = k^\alpha$$

$$\rightarrow c_1 = \frac{1}{1 + \alpha \beta} \cdot k^\alpha = \frac{1}{2} k$$

$$k_2 = \alpha \beta c_1 \rightarrow k_2 = \frac{\alpha \beta}{1 + \alpha \beta} \cdot k^\alpha = \frac{1}{2} k$$

$$\text{或} \quad c_1 + k_2 = k^\alpha \rightarrow k_2 = k^\alpha - c_1 = \frac{\alpha \beta}{1 + \alpha \beta} \cdot k^\alpha$$

$$c_2 = \beta (\alpha k_2^{\alpha-1}) c_1 \rightarrow c_2 = \left(\frac{\alpha \beta}{1 + \alpha \beta} \right)^\alpha \cdot (k^\alpha)^\alpha = \frac{1}{2} k$$

$$\text{或} \quad c_2 = k_2^\alpha \rightarrow c_2 = \left(\frac{\alpha \beta}{1 + \alpha \beta} \cdot k^\alpha \right)^\alpha$$

$$\rightarrow i_1 = k_2 = \frac{\alpha \beta}{1 + \alpha \beta} \cdot k^\alpha = \frac{1}{2} k$$

$$i_2 = 0$$

$$\text{利用資本市場} \quad 1 + r^* = \text{MPK}_2 + 1 - \delta$$

$$\rightarrow r^* = \text{MPK}_2 - \delta$$

$$\rightarrow r^* = \alpha k_2^{\alpha-1} - 1 = \alpha \left(\frac{\alpha \beta}{1 + \alpha \beta} \cdot k^\alpha \right)^{\alpha-1} - 1$$

或利用(假想的)債券市場

$$u'(c_1) = \beta u'(c_2)(1+r)$$

$$\rightarrow \frac{1}{c_1} = \frac{\beta}{c_2}(1+r)$$

$$\rightarrow r = \frac{c_2}{\beta c_1} - 1$$

$$c_2 = \beta(\alpha k_2^{\alpha-1})c_1$$

$$\rightarrow r^* = \frac{\beta(\alpha k_2^{\alpha-1})c_1}{\beta c_1} - 1$$

$$r^* = (\alpha k_2^{\alpha-1}) - 1$$

$$r^* = \alpha \left(\frac{\alpha\beta}{1+\alpha\beta} \cdot k^\alpha \right)^{\alpha-1} - 1 = 0$$

- (3) 計算三期 ($t = 1, 2, 3$) 模型的均衡解：消費 c_1^*, c_2^*, c_3^* 與投資 i_1^*, i_2^*, i_3^* 為何？均衡實質利率 r_1^*, r_2^* 為何？

假設經濟體的資訊如下： $u(c_t) = \ln c_t$ 、 $y_t = A_t k_t^\alpha$ 、 $\alpha = 1$ 、 $A_1 = 1$ 、 $A_2 = 2$ 、 $A_3 = 3$ 、 $\beta = 1$ 、 $\delta = 1$ 。已知資本初始條件 $k_1 = 1$ 。(8%)

$$\max_{c_1, c_2, c_3} \ln c_1 + \beta \ln c_2 + \beta^2 \ln c_3$$

$$\begin{aligned} \text{s.t.} \quad c_1 + k_2 &= A_1 k_1^\alpha \\ c_2 + k_3 &= A_2 k_2^\alpha \\ c_3 &= A_3 k_3^\alpha \end{aligned}$$

帶入本題參數

$$\begin{aligned} \max_{c_1, c_2, c_3} \quad & \ln c_1 + \beta \ln c_2 + \beta^2 \ln c_3 \\ \text{s.t.} \quad & c_1 + k_2 = 1 \\ & c_2 + k_3 = 2k_2 \\ & c_3 = 3k_3 \end{aligned}$$

$$\begin{aligned} \text{FOC} \quad [k_2] \quad & \frac{1}{c_1} = \frac{\beta}{c_2} (2) \quad \rightarrow \quad c_2 = 2\beta c_1 \\ [k_3] \quad & \frac{1}{c_1} = \frac{\beta^2}{c_3} (3)(2) \quad \rightarrow \quad c_3 = 6\beta^2 c_1 \end{aligned}$$

$$\begin{aligned} c_3 = 3k_3 \quad & \rightarrow \quad 6\beta^2 c_1 = 3k_3 \\ & \rightarrow \quad k_3 = 2\beta^2 c_1 \end{aligned}$$

$$\begin{aligned} c_2 + k_3 = 2k_2 \quad & \rightarrow \quad 2\beta c_1 + 2\beta^2 c_1 = 2k_2 \\ & \rightarrow \quad k_2 = (\beta + \beta^2)c_1 \end{aligned}$$

$$\begin{aligned} c_1 + k_2 = 1 \quad & \rightarrow \quad c_1 + (\beta + \beta^2)c_1 = 1 \\ & \rightarrow \quad c_1 = \frac{1}{1 + \beta + \beta^2} = \frac{1}{3} \end{aligned}$$

$$c_2 = 2\beta c_1 \quad \rightarrow \quad c_2 = \frac{2\beta}{1 + \beta + \beta^2} = \frac{2}{3}$$

$$c_3 = 6\beta^2 c_1 \quad \rightarrow \quad c_3 = \frac{6\beta^2}{1 + \beta + \beta^2} = \frac{6}{3} = 2$$

$$k_2 = (\beta + \beta^2)c_1 \quad \rightarrow \quad i_1 = k_2 = \frac{\beta + \beta^2}{1 + \beta + \beta^2} = \frac{2}{3}$$

$$k_3 = 2\beta^2 c_1 \quad \rightarrow \quad i_2 = k_3 = \frac{2\beta^2}{1 + \beta + \beta^2} = \frac{2}{3}$$

利用資本市場

$$1 + r_1^* = \mathbf{MPK}_2 + 1 - \delta$$

$$\rightarrow r_1^* = \mathbf{MPK}_2 - \delta$$

$$\rightarrow r_1^* = 2 - 1 = 1$$

$$1 + r_2^* = \mathbf{MPK}_3 + 1 - \delta$$

$$\rightarrow r_2^* = \mathbf{MPK}_3 - \delta$$

$$\rightarrow r_2^* = 3 - 1 = 2$$

或利用(假想的)債券市場

$$u'(c_1) = \beta u'(c_2)(1 + r_1)$$

$$\rightarrow \frac{1}{c_1} = \frac{\beta}{c_2}(1 + r_1)$$

$$\rightarrow r_1 = \frac{c_2}{\beta c_1} - 1$$

$$c_2 = 2\beta c_1$$

$$\rightarrow r_1^* = \frac{2\beta c_1}{\beta c_1} - 1$$

$$\rightarrow r_1^* = 2 - 1 = 1$$

$$u'(c_2) = \beta u'(c_3)(1 + r_2)$$

$$\rightarrow \frac{1}{c_2} = \frac{\beta}{c_3}(1 + r_2)$$

$$\rightarrow r_2 = \frac{c_3}{\beta c_2} - 1$$

$$c_3 = 3\beta c_2$$

$$\rightarrow r_2^* = \frac{3\beta c_2}{\beta c_2} - 1$$

$$\rightarrow r_2^* = 3 - 1 = 2$$